Quantum field theory I , Homework.

Exercise 1.8

Lagrangian: $$L = \int_0^\ell dx\, \mathcal{L} = \int_0^\ell dx \left[\frac{\mu}{2} \left(\frac{\partial y}{\partial t}\right)^2 - \frac{T}{2}\left(\frac{\partial y}{\partial x}\right)^2 \right]$$

We express $y(x,t)$ as a fourier transform such that

$$y(x,t) = \sqrt{\frac{2}{\ell}} \sum_{n=1}^{\infty} \sin\left(\frac{n\pi x}{\ell}\right) q_n(t)$$

Plugging this into equation for L we get

$$L = \int_0^\ell dx \left[\frac{\mu}{2} \left(\frac{\partial}{\partial t}\left[\sqrt{\frac{2}{\ell}} \sum_{n=1}^{\infty} \sin\left(\frac{n\pi x}{\ell}\right) q_n(t) \right]\right)^2 - \frac{T}{2} \left(\frac{\partial}{\partial x}\left[\dots \right]\right)^2 \right]$$

$$= \int_0^\ell dx \left[\frac{\mu}{2} \cdot \frac{2}{\ell} \left(\sum_{n=1}^{\infty} \sin\left(\frac{n\pi x}{\ell}\right) \dot{q}_n(t) \right)^2 - \frac{T}{2} \cdot \frac{2}{\ell} \left(\sum_{n=1}^{\infty} \frac{n\pi}{\ell} \cos\left(\frac{n\pi x}{\ell}\right) q_n(t) \right)^2 \right]$$

The cross terms in $\left(\sum_{n=1} \sin\left(\frac{n\pi x}{\ell}\right) \dot{q}_n(t) \right)^2$ and $\left(\sum_{n=1}^{\infty} \frac{n\pi}{\ell} \cos\left(\frac{\pi n x}{\ell}\right) q_n(t) \right)^2$

vanish when doing the integral since

$$\int_0^\ell dx \sin\left(\frac{n\pi x}{\ell}\right) \sin\left(\frac{k\pi x}{\ell}\right) = \frac{\ell n \sin(\pi k)\cos(\pi n) - k\ell \cos(\pi k)\sin(\pi n)}{\pi k^2 - \pi n^2} = 0$$

$$- \int_0^\ell dx \cos\left(\frac{n\pi x}{\ell}\right) \cos\left(\frac{k\pi x}{\ell}\right) = \frac{-k\ell \sin(\pi k)\cos(\pi n) + \ell n \cos(\pi k)\sin(\pi n)}{\pi k^2 - \pi n^2} = 0$$

These are zero since $\sin(\pi k) = 0$, $\sin(n\pi) = 0$ for $k, n \in \mathbb{Z}$

this means that under the integral we can write

$$\left(\sum_{n=1}^{\infty} \sin\left(\frac{n\pi x}{\ell}\right) \dot{q}_n(t) \right)^2 \quad \text{as} \quad \sum_{n=1}^{\infty} \sin^2\left(\frac{n\pi x}{\ell}\right) \dot{q}_n^2(t)$$

and $$\left(\sum_{n=1}^{\infty} n \cos\left(\frac{n\pi x}{\ell}\right) q(t) \right)^2 \quad \text{as} \quad \sum_{n=1}^{\infty} n^2 \cos^2\left(\frac{n\pi x}{\ell}\right) q_n^2(t)$$

So we have $$L = \sum_{n=1}^{\infty} \int_0^\ell dx \left[\frac{\mu}{\ell} \sin^2\left(\frac{n\pi x}{\ell}\right) \dot{q}_n(t)^2 - \frac{T}{\ell^3} n^2\pi^2 \cos^2\left(\frac{n\pi x}{\ell}\right) q_n(t)^2 \right]$$

Let's compute the integrals (via calculators since I'm too lazy)

$$\int_0^l dx \quad \sin^2\left(\frac{\pi n x}{l}\right) = \frac{1}{4} l \left(2 - \frac{\sin(2\pi n)}{\pi n}\right) \quad \Big\| \quad \sin(2\pi n) = 0$$

$$= \frac{1}{2} l$$

$$\int_0^l dx \quad \cos^2\left(\frac{\pi n x}{l}\right) = \frac{l}{4\pi n}\left(2\pi n + \sin(2\pi n)\right) \quad \Big\| \quad \sin(2\pi n) = 0$$

$$= \frac{l}{2}$$

Inputting the results into L we get

$$L = \sum_{n=1}^{\infty}\left[\mu/2 \; \dot{q}_n(t)^2 - \frac{T}{2l^2} n^2\pi^2 \, q_n(t)^2\right] \quad \Big\| \quad \text{We define } \omega_n = \sqrt{\frac{T}{\mu}}\,\frac{n\pi}{l}$$

$$\Rightarrow \quad L = \sum_{n=1}^{\infty}\left[\frac{1}{2}\mu\,\dot{q}_n^2 - \frac{1}{2}\mu\omega_n^2 q_n^2\right] \quad \square$$

We have reached the desired form for the Lagrangian.

Exercise 1.9

Hamiltonian density: $\mathcal{H} = \pi^a \dot{y}_a - \mathcal{L} \quad \Big\| \quad \pi^a = \frac{\partial \mathcal{L}}{\partial \dot{y}_a}$

And Hamiltonian is $H = \int d^3x \, \mathcal{H}$

We are working on one-dimensional problem:

$$H = \int dx\, \mathcal{H} = \int_0^l dl \,(\pi_y \dot{y} - \mathcal{L})$$

$$= \left(\int_0^l dl \;\pi_y \dot{y}\right) - L$$

$$\Big\| \quad \dot{y} = \sqrt{\frac{2}{l}}\sum_{n=1}^{\infty}\sin\left(\frac{n\pi x}{l}\right)\dot{q}_n(t)$$

$$\pi_y = \frac{\partial \mathcal{L}}{\partial \dot{y}} = \frac{\partial}{\partial \dot{y}}\left(\frac{\mu}{2}\dot{y}^2\right) = \mu\dot{y}$$

$$= \int_0^l dl\, \mu\, \dot{y}^2 - L$$

using same arguments as in previous exercise we simplify $\dot{y}^2$ term insid integral

$$H = \left[\sum_{n=1}^{\infty}\int_0^l dl \,\frac{2\mu}{l}\sin^2\left(\frac{n\pi x}{l}\right)\dot{q}_n^2(t)\right] - L$$

We use the solution to integral from previous exercise

$$H = \sum_{n=1}^{\infty}\frac{2\mu}{l}\dot{q}_n^2 \cdot \frac{1}{2} l \quad - L$$

We now put the equation for L

$$\Rightarrow \quad H = \sum_{n=1}^{\infty} \mu \dot{q}_n^2 - \sum_{n=1}^{\infty} \left[\frac{1}{2}\mu \dot{q}_n^2 - \frac{1}{2}\mu \omega_n^2 q_n^2 \right]$$

$$H = \sum_{n=1}^{\infty} \left[\frac{1}{2}\mu \dot{q}_n^2 + \frac{1}{2}\mu \omega_n^2 q_n^2 \right]$$

in terms of $p_n = \mu \dot{q}_n$ where p_n is the canonical momentum for n-th coordinate we get:

$$H = \sum_{n=1}^{\infty} \left[\frac{1}{2\mu} p_n^2 + \frac{1}{2}\mu \omega_n^2 q_n^2 \right]$$

The canonical commutation relations are $[q_n, p_m] = i\,\delta_{nm}$ with other commutations being 0 i.e $[q_n, q_m] = 0 = [p_n, p_m]$

This leads directly to $[y(x,t), y(x',t)] = 0$, $[\Pi_y(x,t), \Pi_y(x',t)] = 0$, since $y \sim q_n$ & $\Pi_y \sim p_n$

Now let's calculate $[y(x,t), \Pi_y(x',t)]$

$$y(x,t) = \sqrt{\frac{2}{\ell}} \sum_{n=1}^{\infty} \sin\left(\frac{n\pi x}{\ell}\right) q_n(t)$$

$$\Pi_y(x',t) = \mu \dot{y}(x',t) = \sqrt{\frac{2}{\ell}} \sum_{n=1}^{\infty} \sin\left(\frac{n\pi x}{\ell}\right) p_n(t)$$

$$\Rightarrow \quad [y(x,t), \Pi_y(x',t)] = \frac{2}{\ell}\left(\sum_{n=1}^{\infty} \sin\left(\frac{n\pi x}{\ell}\right) q_n\right)\left(\sum_{n=1}^{\infty} \sin\left(\frac{n\pi x'}{\ell}\right) p_n\right) - \frac{2}{\ell}\left(\sum_{n=1}^{\infty} \sin\left(\frac{n\pi x'}{\ell}\right) p_n\right)\left(\sum_{n=1}^{\infty} \sin\left(\frac{n\pi x}{\ell}\right) q_n\right)$$

$$= \frac{2}{\ell} \sum_{n=1}^{\infty}\sum_{k=1}^{\infty} \sin\left(\frac{n\pi x}{\ell}\right) \sin\left(\frac{k\pi x'}{\ell}\right) \underbrace{[q_n, p_k]}_{= i\,\delta_{nk}}$$

$$= \frac{2}{\ell} i \sum_{n=1}^{\infty} \sin\left(\frac{n\pi x}{\ell}\right) \sin\left(\frac{n\pi x'}{\ell}\right)$$

Using the equations
$$\begin{cases} \cos(x+y) = \cos(x)\cos(y) - \sin(x)\sin(y) \\ \cos(x-y) = \cos(x)\cos(y) + \sin(x)\sin(y) \end{cases}$$

We recover equation for $\sin(x)\sin(y)$:

$$\sin(x)\sin(y) = \frac{1}{2}\left(\cos(x-y) - \cos(x+y)\right)$$

This leads to $[y(x,t), \Pi_y(x',t)] = \frac{i}{\ell} \sum_{n=1}^{\infty} \cos\left(\frac{n\pi}{\ell}(x-x')\right) - \cos\left(\frac{n\pi}{\ell}(x+x')\right)$

The sum for the case $n=0$ is zero so we can write

$$[y(x,t), \Pi_y(x',t)] = \frac{i}{\ell} \sum_{n=0}^{\infty} \cos\left[\frac{n\pi}{\ell}(x-x')\right] - \cos\left[\frac{n\pi}{\ell}(x+x')\right]$$

Using equation provided we simplify this to:

$$[y(x,t), \Pi_y(x',t)] = i\,\delta(x-x')$$

So we have shown $[y(x,t), y(x',t)] = 0 = [\Pi_y(x,t), \Pi_y(x',t)]$ &

$$[y(x,t), \Pi_y(x',t)] = i\,\delta(x-x')$$

Exercise 1.10

Lagrangian density:

$$\mathcal{L} = \frac{1}{2}\partial_\mu\phi\,\partial^\mu\phi + \frac{1}{2}\kappa(\partial_\mu\phi\partial^\mu\phi)^2 - \frac{1}{2}m^2\phi^2 - g\phi^3 + \lambda\phi^4 - \sigma\phi^6$$

(a) System in 3 space dimension $\Rightarrow$ 4D - spacetime

Action is dimensionless : $[d^4x] = -4 \Rightarrow [\mathcal{L}] = 4$

Going term by term:

$[x^\mu] = -1 \Rightarrow [\partial_\mu] = [\partial^\mu] = -1 \Rightarrow [\partial_\mu\phi\,\partial^\mu\phi] = 2[\partial_\mu] + 2[\phi]$

$4 = -2 + 2[\phi]$

We know $[\partial_\mu\phi\partial^\mu\phi] = [\mathcal{L}] = 4$

$\Rightarrow 2[\phi] = 4 + 2 = 6$

$\Rightarrow \underline{[\phi] = 3}$

we also have $[\kappa(\partial_\mu\phi\partial^\mu\phi)^2] = 4 = [\kappa] + 2[\partial_\mu\phi\partial^\mu\phi] = [\kappa] + 2\cdot4$

$\Rightarrow \underline{[\kappa] = -4}$

$[m^2\phi^2] = 4 \Rightarrow 2[m] + 2[\phi] = 4 \Rightarrow 2[m] = -2 \Rightarrow \underline{[m] = -1}$

$[g\phi^3] = 4 \Rightarrow \underline{[g] = -5}$, $[\lambda\phi^4] = 4 \Rightarrow \underline{[\lambda] = -8}$

$[\sigma\phi^6] = 4 \Rightarrow \underline{[\sigma] = -14}$

The mass dimensions are $[\phi] = 3$, $[\kappa] = -4$, $[m] = -1$, $[g] = -5$, $[\lambda] = -8$ and $[\sigma] = -14$ for 3 space dimensions

b) For 2 space dimensions we have 3D - spacetime

$[S] = 0 \Rightarrow [\mathcal{L}] = 3$ & $[\partial_\mu] = -1$ & $[\partial_\mu\phi\partial^\mu\phi] = 3 \Rightarrow \underline{[\phi] = \frac{5}{2}}$

$[\kappa(\partial_\mu\partial\phi\partial^\mu\phi)^2] = [\kappa] + 2\cdot3 = 3 \Rightarrow \underline{[\kappa] = -3}$

$[m^2\phi^2] = 3 \Rightarrow 2[m] = 3 - 2[\phi] = -2 \Rightarrow \underline{[m] = -1}$

$[g\phi^3] = 3 \Rightarrow [g] = 3 - 3\cdot\frac{5}{2} = 3 - \frac{15}{2} = -\frac{9}{2} \Rightarrow [g] = -9/2$

$[\lambda\phi^4] = 3 \Rightarrow [\lambda] = 3 - 4\cdot\frac{5}{2} = -7 \Rightarrow [\lambda] = -7$

$[\sigma\phi^6] = 3 \Rightarrow [\sigma] = 3 - 6\cdot\frac{5}{2} = -12 \Rightarrow [\sigma] = -12$

The mass dimensions are $[\phi] = \frac{5}{2}$, $[\kappa] = -3$, $[m] = -1$, $[g] = -\frac{9}{2}$, $[\lambda] = -7$ and $[\sigma] = -12$ for 2 space dimensions.

c) 5 spatial dimensions $\Rightarrow$ 6D- spacetime

$[S] = 0 \Rightarrow [\mathcal{L}] = 6$, $[\partial_\mu] = -1$

$[\partial_\mu\phi\partial^\mu\phi] = 6 \Rightarrow 2[\phi] = 8 \Rightarrow [\phi] = 4$

$[\kappa(\partial_\mu\phi\partial^\mu\phi)^2] = 6 \Rightarrow [\kappa] = 6 - 2\cdot 6 = -6 \Rightarrow [\kappa] = -6$

$[m^2\phi^2] = 6 \Rightarrow 2[m] = 6 - 8 = -2 \Rightarrow [m] = -1$

$[g\phi^3] = 6 \Rightarrow [g] = 6 - 3\cdot 4 = -6 \Rightarrow [g] = -6$

$[\lambda\phi^4] = 6 \Rightarrow [\lambda] = 6 - 4\cdot 4 = -10 \Rightarrow [\lambda] = -10$

$[\sigma\phi^6] = 6 \Rightarrow [\sigma] = 6 - 6\cdot 4 = -18 \Rightarrow [\sigma] = -18$

The mass dimensions are $[\phi] = 4$, $[\kappa] = -6$, $[m] = -1$, $[g] = -6$, $[\lambda] = -10$, $[\sigma] = -18$ for 5 space dimensions